

Research Methods & Assessment

*Comprehensive EPPP Preparation Study Guide
For Destynee | Dr. DeDe's Concept List*

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1. Reliability

Definition: Consistency/stability of measurement; extent to which test yields same results on repeated administration.

Types: - **Test-Retest Reliability:** Consistency across time - **Inter-rater Reliability:** Consistency across different raters/observers - **Internal Consistency:** Consistency across items (Cronbach's alpha, split-half) - **Parallel Forms Reliability:** Consistency between equivalent forms of test

Factors Affecting: Test length, item homogeneity, test-retest interval, testing conditions.

Source: Anastasi, A., & Urbina, S. (1997). *Psychological testing* (7th ed.). Prentice Hall.

2. Validity

Definition: Extent to which a test measures what it claims to measure.

Types: - **Content Validity:** Items represent domain being measured - **Criterion Validity:** Correlation with external criterion - Concurrent: Correlation with present criterion - Predictive: Predicts future criterion - **Construct Validity:** Measures theoretical construct (convergent, discriminant) - **Face Validity:** Appears to measure what it should (superficial)

Relationship with Reliability: A test can be reliable without being valid, but cannot be valid without being reliable.

Source: Messick, S. (1995). Validity of psychological assessment. *American Psychologist*.

3. Correlation Coefficient

Definition: Statistical measure (r) of the strength and direction of linear relationship between two variables.

Range: -1.00 to +1.00 - 0 = no relationship - ± 1.00 = perfect relationship

Interpretation: - Small: ± 0.10 to ± 0.30 - Medium: ± 0.30 to ± 0.50 - Large: ± 0.50 and above

Important: Correlation $\neq$ Causation

Coefficient of Determination (r^2): Proportion of variance in one variable accounted for by another.

Source: Cohen, J. (1988). *Statistical power analysis for the behavioral sciences* (2nd ed.). Lawrence Erlbaum.

4. Cross-Sectional Study

Definition: Research design comparing different age groups at a single point in time.

Pros: - Quick and inexpensive - No participant attrition - Large samples possible

Cons: - Cohort effects confounded with age effects - Cannot study individual change - Selection bias

Best For: Descriptive data, preliminary research.

Source: Schaie, K. W. (1986). *Adult development and aging*. Little, Brown.

5. Longitudinal Study

Definition: Research design following same individuals over time to study developmental changes.

Pros: - Can study individual change - No cohort effects - Establishes temporal sequence

Cons: - Time-consuming and expensive - Participant attrition (selective dropout) - Practice effects - Historical changes during study period

Source: Schaie, K. W. (1986). *Adult development and aging*. Little, Brown.

6. Microgenetic Study

Definition: Research design tracking the process of change as it occurs by making frequent observations over a period of rapid development or transition.

Characteristics: - Intensive, frequent measurements - Captures transitions in real-time - Examines mechanisms of change - Shorter timeframe than longitudinal

Uses: - Understanding how new strategies emerge - Identifying factors that promote change - Tracking developmental transitions

Source: Siegler, R. S., & Crowley, K. (1991). The microgenetic method. *American Psychologist*, 46(6), 606-620.

7. Cohort Effects

Definition: Differences in outcomes between groups born at different times due to different historical/social experiences.

Which Study Type: Most commonly observed in **cross-sectional studies** where different age groups are compared at one point in time.

Example: Comparing depression rates in 20-year-olds vs. 60-year-olds in 2020 confounds age effects with cohort effects (different historical experiences: Great Depression vs. COVID-19 pandemic).

Solutions: Longitudinal designs, sequential designs.

Source: Schaie, K. W. (1965). A general model for the study of developmental problems. *Psychological Bulletin*.

8. Habituation

Definition: Decreased response to repeated stimulus; indicates learning and recognition.

Why Important: - Used to study infant perception and memory - Indicates information processing - Measures discrimination between stimuli - Non-verbal assessment method

Dishabituation: Recovery of attention when novel stimulus introduced; indicates discrimination between familiar and novel.

Source: Fantz, R. L. (1964). Visual experience in infants: Decreased attention to familiar patterns relative to novel ones. *Science*.

9. ADLs vs IADLs

ADLs (Activities of Daily Living): Basic self-care tasks - Bathing/showering - Dressing - Toileting - Transferring (mobility) - Continence - Feeding

IADLs (Instrumental Activities of Daily Living): Complex tasks for independent living - Meal preparation - Medication management - Financial management - Transportation/shopping - Housekeeping - Phone/technology use

Assessment Tools: Katz ADL Scale, Lawton IADL Scale.

Source: Katz, S., et al. (1963). Studies of illness in the aged. *JAMA*; Lawton, M. P., & Brody, E. M. (1969). Assessment of older people. *The Gerontologist*.

10. Learning Disabilities - Diagnosis Problems

Major Issues with Current LD Diagnosis:

Discrepancy Model Problems: - Delays identification (wait for gap between IQ and achievement) - Doesn't account for comorbidities - Arbitrary cutoffs

Other Issues: - Heterogeneity within categories - Over-identification and under-identification concerns - High comorbidity with ADHD - Lack of clear biological markers

Alternative Approach: Response to Intervention (RTI): Identify through failure to respond to evidence-based instruction; reduces over-identification.

Source: Fletcher, J. M., et al. (2019). The IQ-achievement discrepancy classification approach. *Contemporary School Psychology*.

11. Three Approaches to Defining Learning Differences

1. IQ-Achievement Discrepancy: - Significant difference between IQ and academic achievement - Criticized for delaying services

2. Response to Intervention (RTI): - Failure to respond to evidence-based instruction - Early identification focus - Tiered intervention system

3. Pattern of Strengths and Weaknesses: - Intra-individual differences in cognitive abilities - Specific deficits in processing areas - Relative strengths in other areas

Source: Flanagan, D. P., & Alfonso, V. C. (2017). *Essentials of specific learning disability identification* (2nd ed.). Wiley.

12. Youngstrom's 3 P's of Evidence-Based Assessment

Three P's Framework:

1. Problem: What is the presenting problem? - Diagnostic hypothesis generation - Differential diagnosis

2. Predict: What will happen? - Prognosis prediction - Risk assessment - Treatment planning

3. Plan: What should we do? - Treatment selection - Progress monitoring - Outcome evaluation

Source: Youngstrom, E. A. (2008). Evidence-based strategies for the assessment of developmental psychopathology. In R. G. Steele et al. (Eds.), *Handbook of evidence-based therapies for children and adolescents*. Springer.

13. Standardized Assessment

Norm-Referenced Tests: Compare individual to representative sample (norms). - Age norms, grade norms, percentile ranks - Standard scores (mean 100, SD 15 for IQ)

Criterion-Referenced Tests: Compare performance to predetermined standard. - Mastery testing - Curriculum-based measurement

Standardization: Uniform administration and scoring procedures.

Types of Scores: - Raw scores - Standard scores (z-scores, T-scores, IQ scores) - Percentile ranks - Age/grade equivalents

Source: Sattler, J. M. (2018). *Assessment of children: Behavioral, social, and clinical foundations* (6th ed.). Author.

14. Intelligence Testing

Wechsler Scales (WISC, WAIS, WPPSI): - Verbal Comprehension Index - Perceptual Reasoning/Visual Spatial Index - Working Memory Index - Processing Speed Index - Full Scale IQ

Stanford-Binet (SB-5): - Fluid Reasoning - Knowledge - Quantitative Reasoning - Visual-Spatial Processing - Working Memory

Key Concepts: - g factor (general intelligence) - Crystallized vs. Fluid intelligence (Cattell-Horn-Carroll theory)

Source: Wechsler, D. (2014). *Wechsler Intelligence Scale for Children* (5th ed.). Pearson.

15. Triarchic Theory of Intelligence (Sternberg)

Definition: Three distinct types of intelligence.

1. Analytical (Componential): - Abstract reasoning, problem-solving - Academic ability - Correlates with g factor

2. Creative (Experiential): - Dealing with novel situations - Automaticity - Insight and creativity

3. Practical (Contextual): - Adapting to environment - Shaping environment - Selecting environments

Research Support: Creative and practical intelligence predict success beyond IQ; practical intelligence shows cultural specificity; validated across multiple cultures.

Source: Sternberg, R. J. (1985). *Beyond IQ: A triarchic theory of human intelligence*. Cambridge University Press.

16. Theory of Multiple Intelligences (Gardner)

Definition: Eight relatively independent intelligences.

Eight Intelligences: 1. Linguistic 2. Logical-mathematical 3. Musical 4. Spatial 5. Bodily-kinesthetic 6. Interpersonal 7. Intrapersonal 8. Naturalistic

Research Support: Limited empirical validation; popular in education but criticized for lack of psychometric support; not supported by factor analysis research.

Source: Gardner, H. (1983). *Frames of mind: The theory of multiple intelligences*. Basic Books.

17. Flynn Effect

Definition: Substantial and long-sustained increase in both fluid and crystallized intelligence test scores measured in many parts of the world over the 20th century.

Magnitude: Approximately 3 IQ points per decade.

Possible Causes: - Improved nutrition - Better healthcare - Increased education - Smaller families/more parental attention - Environmental complexity - Test-taking familiarity

Reverse Flynn Effect: Recent evidence suggests decline in some countries.

Source: Flynn, J. R. (1984). The mean IQ of Americans: Massive gains 1932 to 1978. *Psychological Bulletin*.

18. Five Factor Model of Personality

Whose Model: Costa & McCrae (primary developers); building on earlier lexical research

Five Factors (OCEAN):

1. **Openness to Experience:** Creativity, curiosity, preference for variety

2. **Conscientiousness:** Organization, dependability, self-discipline

3. **Extraversion:** Sociability, energy, positive emotions

4. **Agreeableness:** Trust, altruism, cooperation

5. **Neuroticism:** Emotional instability, anxiety, negative affect (reverse: Emotional Stability)

Assessment: NEO-PI-R, NEO-PI-3

Source: Costa, P. T., Jr., & McCrae, R. R. (1992). *Revised NEO Personality Inventory*. Psychological Assessment Resources.

19. Behavioral Assessment

Types:

Direct Observation: Observing behavior in natural or controlled settings. - ABC recording (Antecedent-Behavior-Consequence) - Time sampling (interval, momentary) - Event recording

Rating Scales: Standardized scales completed by informants. - BASC-3, CBCL, TRF

Functional Behavioral Assessment (FBA): - Identify function of behavior (attention, escape, sensory, tangible) - Develop intervention based on function

Self-Monitoring: Individual records own behavior.

Source: Kazdin, A. E. (2013). *Behavior modification in applied settings* (7th ed.). Waveland Press.

20. Projective Tests

Rorschach Inkblot Method: - 10 inkblots - Exner Comprehensive System - Cognitive and emotional processing

Thematic Apperception Test (TAT): - Ambiguous pictures - Narrative analysis - Motivation, needs, conflicts

Sentence Completion: - Stem completion tasks - Project personality dynamics

Controversy: Mixed psychometric support; low reliability/validity concerns; still used clinically.

Source: Meyer, G. J., & Viglione, D. J. (2008). An introduction to Rorschach assessment. In R. P. Archer & S. R. Smith (Eds.), *Personality assessment*. Routledge.

21. Cultural Considerations in Assessment

Cultural Test Bias: - Content bias - Linguistic bias - Norm sample bias

Best Practices: - Use culturally appropriate norms - Consider acculturation - Language proficiency assessment
- Dynamic assessment - Non-biased assessment tools

Acculturative Stress: Psychological impact of adapting to new culture; affects test performance and mental health.

Source: Rhodes, R. L., et al. (2005). *Assessing culturally and linguistically diverse students*. Guilford Press.